Carbon Fibre Hydrogen Tank

CRAFT - Carbon Reinforced Advanced Fuel Tank

By Fatima, Marco, Matthew, Lucas, Cheryl

Rationale/Context

For the UK to become net zero carbon by 2050, we need alternate fuel sources for vehicles. As hydrogen presents itself as a really good alternative, we must consider the material of the storage vessel that lets us best implement hydrogen fuel technology.

Functions

Must hold a sizable amount of pressure.

Must be safe 'yield before break' philosophy.

Free variables

Thickness of tank walls

Holdable pressure

Weight/mass

Constraints

Length of cylinder **400mm**

Outer Radius **50mm**

Material index m_1

Let σ_y equal to σ_θ, that is:

$$\sigma_y \leq \frac{pb}{t}$$

Set σ_y less than or equal to σ_{fr}, that is:

$$\sigma_y \leq \frac{CK_{1C}}{\sqrt{\pi a}}$$

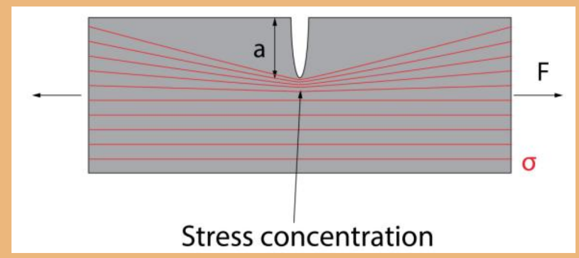
Rearrange to make a the subject

$$a \leq \frac{C^2}{\pi}\frac{{K_{1C}}^2}{{\sigma_y}^2}$$

The first material index, m_1 is therefore:

$$m_1 = \frac{{K_{1C}}^2}{{\sigma_y}^2}$$

Material index m_2



$$\sigma_{fr} = \frac{CK_{1C}}{\sqrt{\pi \frac{t}{2}}}$$

Using this rearrangement

$$t = \frac{pb}{\sigma_y}$$

We get after substitution

$$\sigma_y \leq \frac{CK_{1C}}{\sqrt{\pi \frac{pb}{2\sigma_y}}}$$

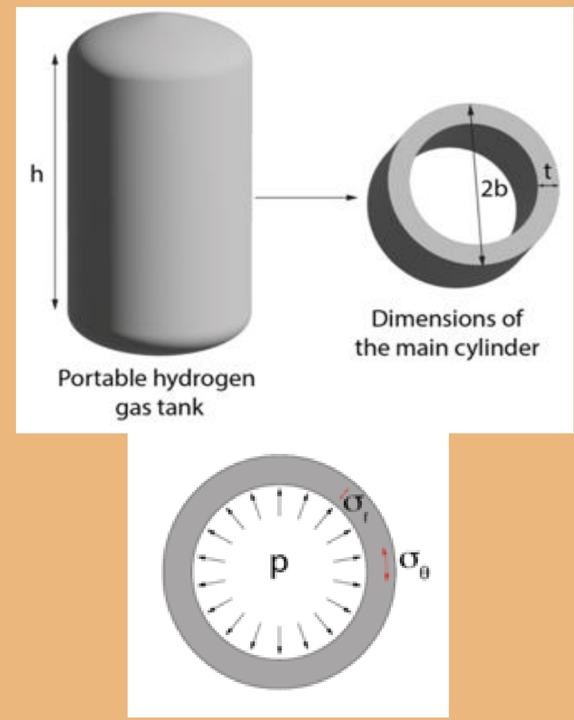
Rearrang for pressure

$$p \leq C^2 \frac{2}{\pi b}\frac{{K_{1C}}^2}{\sigma_y}$$

Giving us another material index

$$m_2 = \frac{{K_{1C}}^2}{\sigma_y}$$

Material index m_3



$$m = (2\pi bt)h\rho$$

Substitute as with m_2

$$m \geq 2\pi b \frac{pb}{\sigma_y} h\rho$$

$$m \geq (p)(2\pi b^2)\frac{\rho}{\sigma_y}$$

Giving us another material index

$$m_3 = \frac{\sigma_y}{\rho}$$

Design Objectives:

1. High fracture toughness to yield ratio
2. Resistant to hydrogen embrittlement
3. High yield stress to density ratio
4. Flexibility (ability to shape into cylinder)
5. Reduced environmental impact

Why Pro?

We believe to make an impact we need to consider how sustainable our solution is in the long run, which means a more expensive but more robust solution.

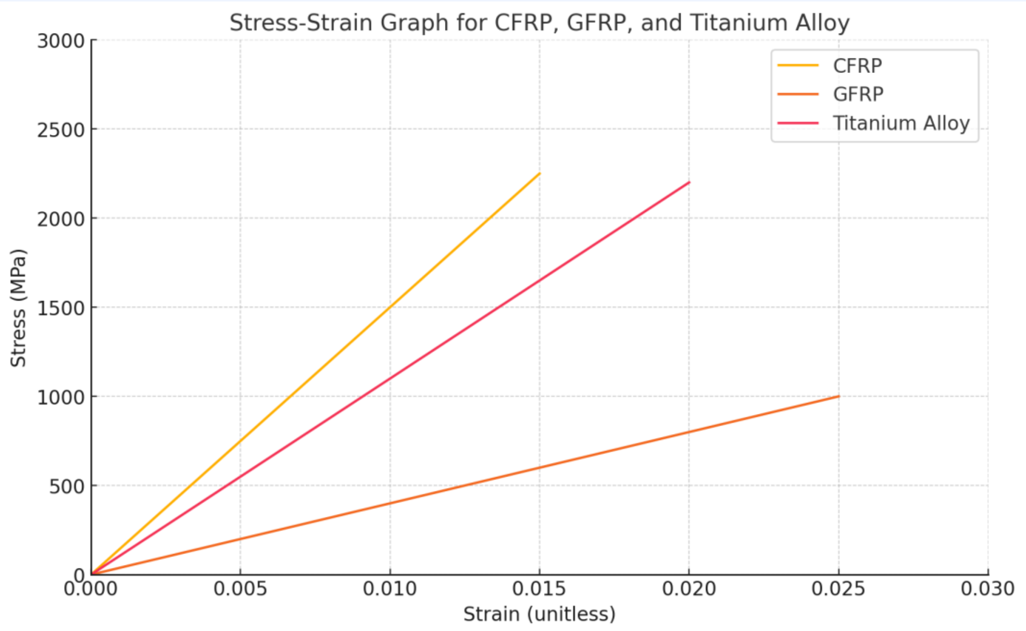
Equation terms

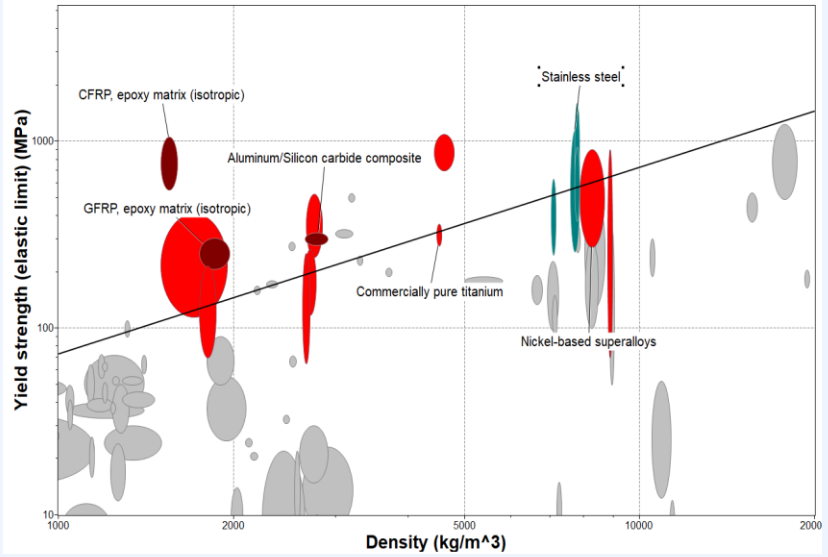
σ_y: Yield stress ($N \cdot m^{-2}$)
σ_{fr}: Fracture stress ($N \cdot m^{-2}$)
C: Constant
K_{1C}: Plane Strain Fracture toughness ($N \cdot m^{-\frac{3}{2}}$)
a: Internal half crack length (m)
t: Wall thickness (m)
b: outer diameter (m)
p: pressure ($N \cdot m^{-2}$)
h: Length of storage tank (m)
ρ: Density ($kg \cdot m^{-3}$)
m=mass (kg)

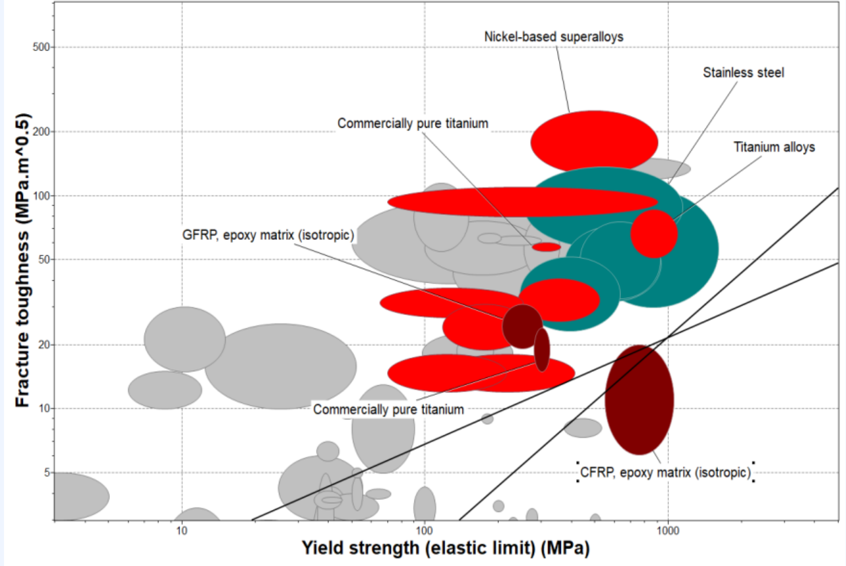
Materials Analysis

Material Shortlist	Pure Titanium, Nickel-based Superalloys, Stainless Steel, Titanium Alloys, CFRP, GFRP and AlSiC
Thermal Properties	Eliminates Stainless Steel as it has the highest thermal conductivity which poses a safety issue.
Durability	Eliminates Pure Titanium, as pure, more prone to hydrogen embrittlement, leading to failure quicker than other metals and composites.
Environmental Impact	Eliminates AlSiC as it is an energy intensive process to extract and manufacture and cannot be recycled effectively.
Density	Eliminates Nickel-based Superalloys as has the highest density of all remaining materials which isn't very desirable for companies.
Final Material Shortlist	Titanium Alloys, CFRP, GFRP

Mechanical Properties	[1]
Young's Modulus (GPa)	294
Strength to Weight ratio (Mpa/g/cm^3)	3026
Tensile Strength parallel to fibres (MPa)	5407
Fracture Toughness (KIC)	0.5- 1.2
Thermal Properties	
Thermal Conductivity parallel to fibres (W/m·K)	5-25
Thermal Conductivity perpendicular to fibres (W/m·K)	0.5-1.5
General Properties	
Density (g/cm^3)	1.79
Durability (years)	15
Cost (£/Kg)	16-25
Processing Energy (MJ/Kg)	200-300







How long does the tank need to last ?

7,400 miles per year [12]

***Estimated:** 20 - 27 years (150k – 200k miles)* [13]

***Realistic:** 3–5 years (22k - 37k miles) (58% UK)* [14]

Consider: Electric vehicles are not desirable because of poor batteries; a reliable storage technology is necessary for an ICE alternative

Extraction

Titanium:
- Extracted from Ti ore
- Conversion to TiO_2
- Kroll Process [4]

Carbon Fibre:
- Heat polymer precursor [2] to remove non-carbon atoms
- Surface treatments

Glass Fibre:
- Raw materials mined from quarries
- Melt to form molten glass
- Extrude through fine nozzles -> create thin filaments [3]

Processing

Ti Alloy:
- Purification of Ti sponge
- Melt sponge in vacuum arc remelting furnace to form alloy
- Reforming of alloy [5]

Carbon Fibre:
- Combine CF. with resin matrix (e.g. epoxy) to form prepreg sheets [6, 11]

Glass Fibre:
- GF gets impregnated with resin matrix (e.g., polyester, epoxy) [7]

Manufacturing

Ti Alloy:
- Shaped into cylindrical shape [8, 10]
- Surface treatment to reduce hydrogen embrittlement [9]
- Assemble parts together

Carbon Fibre:
- Formation of cylindrical tank
- Take place in autoclave to solidify structure [11]
- Assemble parts together

Glass Fibre:
- Similarly, CFRP, but wounding take place at room temperature [7]

Durability

Titanium:
- Susceptible to hydrogen embrittlement
- Able to yield before failure
- Most Dense

Carbon Fibre:
- Lasts 15 years with correct servicing [15]
- Brittle
- Least Dense

Glass Fibre:
- Weaker than Carbon Fibre

Maintenance

Titanium:
- Hydrogen Embrittlement (although not common)
- Formation of microcracks causing loss in strength
- Repairs are inexpensive and simple

Carbon Fibre & Glass Fibre:
- Reapplication of topcoat of resin
- Replacing layers or sections [16]
- Repairs are slightly more complex

- **All tanks should undergo regular leak testing**
- **Problems in either tank can be seen visually or with ultrasonic inspection**

Disposal & Recycling

Titanium:
- Large amount of CO_2 are required [17]
- Fairly easy and common to recycle [17]

Carbon Fibre & Glass Fibre:
- Composites are very difficult to recycle [18]
- Specialist facilities are required [18]
- Non-Biodegradable

Material	Extraction, Processing & Manufacturing	Maintenance	Durability	Disposal & Recycling	Environmental Factor	Ranking
Ti Alloy	3	1	2	1	3	2
CFRP	2	2	1	2	3	1
GFRP	1	3	3	3	1	3

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